

AI SQL Agent - Project Notes

Development Notes & Key Learnings

Project Overview

AI SQL Agent is an AI-powered assistant that converts natural language into executable MySQL and MongoDB queries. It integrates multiple AI providers and offers validation, automatic repair, analytics, visualization, and export capabilities.

Project Objectives

Build an intelligent database assistant, simplify database querying, support multiple AI providers, and provide a modular, scalable architecture.

Technologies Used

Python, Streamlit, MySQL, MongoDB, Ollama, OpenAI, Anthropic, Pandas, SQLParse, OpenPyXL and ReportLab.

Major Modules

AI Chat, Provider Manager, Agent Orchestrator, SQL Agent, Mongo Agent, Validators, Repair Agents, Query Service, Schema Explorer, Dashboard Analytics, Query History, Saved Queries and Export.

Development Highlights

Implemented natural language query generation, SQL and MongoDB validation, automatic query repair, execution metrics, interactive charts, PDF/CSV/Excel export, and professional documentation.

Challenges & Solutions

Handled AI-generated invalid queries using validators and repair agents. Added execution metrics, modular architecture, and provider abstraction for maintainability.

Key Learnings

Improved skills in Python, Object-Oriented Programming, Streamlit, Prompt Engineering, SQL, MongoDB, software architecture, Git, and AI integration.

Future Improvements

Add PostgreSQL support, REST API, Docker, cloud deployment, authentication, multiple database connections, and AI chat memory.

Conclusion

The project demonstrates practical AI integration with databases using a clean, modular architecture and serves as a strong portfolio and learning project.